MATH 11 — ELEMENTARY STATISTICS

§8.2

Confidence Intervals for a Population Proportion

Using $\hat{p}$ to estimate p

Today's Goal

After completing this section, you should be able to:

- Verify the requirements for constructing a confidence interval for a proportion
- Calculate a confidence interval for a population proportion
- Interpret a confidence interval for a proportion in context
- Calculate the minimum sample size needed to estimate a population proportion

Learning Target 13

LT.13 Confidence Intervals

I can determine an appropriate confidence interval estimate for the proportion, mean, and standard deviation of a population and I can interpret what the confidence interval means in context.

Requirements

1. The sample must be a simple random sample.
2. The conditions for a binomial probability distribution are satisfied.
    - There is a fixed number of trials.
    - Each trial must have all outcomes classified into exactly two categories: successes and failures.
    - The trials must be independent of one another.
3. There are at least five successes and five failures. This verifies that $n\hat{p} \geq 5$ and $n\hat{q} \geq 5$, so the normal distribution serves as a suitable approximation to the binomial distribution.

Constructing the Confidence Interval

Procedure 8.2.1 Constructing a Confidence Interval for p

1. Identify the values of $\hat{p}$, $\hat{q} = 1 - \hat{p}$, and n.
2. Verify that the requirements listed above are met.
3. Identify the confidence level and find the corresponding critical value $z_{\alpha/2}$.

Constructing the Confidence Interval

Procedure 8.2.1 Constructing a Confidence Interval for p, continued

4. Find the margin of error:

$$E = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

5. The confidence interval is

$$\hat{p} - E < p < \hat{p} + E.$$

The confidence interval can also be expressed in interval notation as $(\hat{p} - E,\ \hat{p} + E)$ or as $\hat{p} \pm E$.

Using Excel for Critical Values

Procedure 8.2.2 Using Excel for Proportion CI Critical Values

To find the critical value $z_{\alpha/2}$ for a confidence interval, use the `NORM.INV` function:

```
=NORM.INV(1-alpha/2, 0, 1)
```

- For 90% confidence ($\alpha = 0.10$): `=NORM.INV(0.95, 0, 1)` returns 1.6449.
- For 95% confidence ($\alpha = 0.05$): `=NORM.INV(0.975, 0, 1)` returns 1.96.
- For 99% confidence ($\alpha = 0.01$): `=NORM.INV(0.995, 0, 1)` returns 2.5758.

Students at the Library

A university library randomly sampled 160 students. Of these 160 students, 104 said they visit the library every week. Construct a 90% confidence interval for the proportion of students at the university who visit the library every week. Round the confidence interval bounds to three decimal places.

1. We are given that $n = 160$ and $\hat{p} = \frac{104}{160} = 0.65$. So $\hat{q} = 1 - 0.65 = 0.35$.

Students at the Library

2. We will assume that the sample is a simple random sample. The conditions for a binomial probability distribution are met.
   - There is a fixed number of trials: $n = 160$.
   - Success is when a student visits the library every week and failure is when they do not.
   - We will assume that sampling is with replacement (so the trials are independent) since the sample size is less than 5% of the population.

There are $n\hat{p} = 104$ successes and $n\hat{q} = 56$ failures, both of which are at least 5.

Students at the Library

3. The confidence level is 90% so $1 - \alpha = 0.90$, which means $\alpha = 0.10$. Table A-1 gives us $z_{0.05} = 1.6449$.

4. The margin of error is

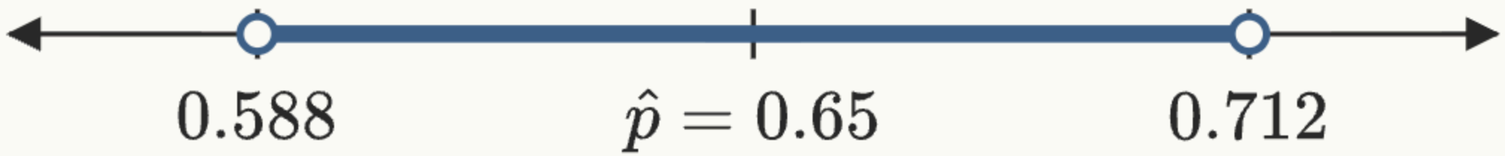
$$\begin{aligned} E &= z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} \\ &= 1.6449\sqrt{\frac{(0.65)(0.35)}{160}} \\ &\approx 0.0620. \end{aligned}$$

5. The confidence interval for p is

$$\begin{aligned} \hat{p} - E < p < \hat{p} + E \\ 0.65 - 0.0620 < p < 0.65 + 0.0620 \\ 0.588 < p < 0.712. \end{aligned}$$

Students at the Library

$$0.588 < p < 0.712$$



Interpretation: If we were to repeat this sampling process many times, 90% of the resulting confidence intervals would contain the true proportion of students at this university who visit the library every week. Based on this sample, the interval $(0.588, 0.712)$ is one such interval.

Changing the Sample Size

The library sampled 160 students. Suppose it had sampled 640 students instead and still found $\hat{p} = 0.65$. It builds a 90% confidence interval again.

Is the new interval wider or narrower?

Home Gardens

In a poll of $2{,}200$ adults, 37% of the respondents have a home garden. Use this to find the 95% confidence interval for the proportion of all adults who have a home garden. Round the confidence interval bounds to three decimal places.

1. We are given that $n = 2{,}200$, $\hat{p} = 0.37$, and $\hat{q} = 0.63$.

Home Gardens

2. We will assume that the sample is a simple random sample. The conditions for a binomial probability distribution are met.
   - There is a fixed number of trials: $n = 2{,}200$.
   - Success is when an adult has a home garden and failure is when they do not.
   - We will assume that sampling is with replacement (so the trials are independent) since the sample size is less than 5% of the population.

There are $n\hat{p} = (2{,}200)(0.37) = 814$ successes and $n\hat{q} = (2{,}200)(0.63) = 1{,}386$ failures, both of which are at least 5.

Home Gardens

3. The confidence level is 95% so $1 - \alpha = 0.95$, which means $\alpha = 0.05$. Table A-1 gives us $z_{0.025} = 1.96$.

4. The margin of error is

$$\begin{aligned} E &= z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} \\ &= 1.96\sqrt{\frac{(0.37)(0.63)}{2{,}200}} \\ &\approx 0.0202. \end{aligned}$$

5. The confidence interval for p is

$$\begin{aligned} \hat{p} - E < p < \hat{p} + E \\ 0.37 - 0.0202 < p < 0.37 + 0.0202 \\ 0.350 < p < 0.390. \end{aligned}$$

Home Gardens

$$0.350 < p < 0.390$$

Interpretation: If we were to repeat this sampling process many times, 95% of the resulting confidence intervals would contain the true proportion of all adults who have a home garden. Based on this sample, the interval $(0.350, 0.390)$ is one such interval.

Sample Size for Estimating a Proportion

Before collecting data, researchers often need to determine how large a sample they need.

Procedure 8.2.5 Sample Size for Estimating a Proportion (with Prior $\hat{p}$)

If a reasonable point estimate $\hat{p}$ is known (from a previous study):

$$n = \frac{\left(z_{\alpha/2}\right)^2 \hat{p}(1-\hat{p})}{E^2}$$

Important: Always round the result **up** to the nearest whole number, even if the decimal is less than 0.5.

Sample Size for Estimating a Proportion

Procedure 8.2.6 Sample Size for Estimating a Proportion (no Prior $\hat{p}$)

If a reasonable point estimate $\hat{p}$ is not known:

$$n = \frac{\left(z_{\alpha/2}\right)^2 (0.25)}{E^2}$$

Important: Always round the result **up** to the nearest whole number.

The formula when $\hat{p}$ is unknown uses 0.25 because the product $\hat{p}(1-\hat{p})$ is maximized when $\hat{p} = 0.5$, giving $0.5 \times 0.5 = 0.25$. This ensures that the sample size will be large enough regardless of what the true proportion turns out to be.

Recycling at Home

A survey of randomly selected U.S. adults showed that 64% of the respondents recycle at home. If we want to conduct a new survey to determine whether that percentage has changed, how many adults must be surveyed in order to be 95% confident that the sample percentage is in error by no more than four percentage points? Round up to the nearest whole number.

(a) Assume that 64% of adults recycle at home, based on the previous survey.

(b) Assume that we have no prior information suggesting a possible point estimate for p.

Recycling at Home

(a) Assume that 64% of adults recycle at home, based on the previous survey.

The confidence level is 95% so $1 - \alpha = 0.95$, which means that $\alpha = 0.05$. Table A-1 gives us $z_{0.025} = 1.96$. We are given that the margin of error is $E = 0.04$.

$$n = \frac{\left(z_{\alpha/2}\right)^2 \hat{p}(1-\hat{p})}{E^2} = \frac{(1.96)^2(0.64)(0.36)}{(0.04)^2} \approx 553.19$$

Rounding up, we need a sample of size $\mathbf{554}$.

Recycling at Home

(b) Assume that we have no prior information for p.

We don't have a value of $\hat{p}$ to use, so we use the formula in Procedure 8.2.6:

$$n = \frac{\left(z_{\alpha/2}\right)^2 (0.25)}{E^2} = \frac{(1.96)^2 (0.25)}{(0.04)^2} \approx 600.25$$

Rounding up, we need a sample of size $\mathbf{601}$.

Notice that when we don't have a prior estimate for $\hat{p}$, the required sample size is larger. This makes sense because we need a larger sample to compensate for the additional uncertainty.

Changing the Confidence Level

A survey of 400 randomly selected residents finds $\hat{p} = 0.52$.

(a) Find the 90% confidence interval.

$$E = 1.6449\sqrt{\frac{(0.52)(0.48)}{400}} \approx 0.0411\text{, so } (0.479,\ 0.561)$$

(b) Find the 99% confidence interval.

$$E = 2.5758\sqrt{\frac{(0.52)(0.48)}{400}} \approx 0.0643\text{, so } (0.456,\ 0.584)$$

(c) Which interval is wider, and why?

The 99% interval. A higher confidence level has a larger critical value, so the margin of error is larger.

Changing the Sample Size

At 95% confidence with $\hat{p} = 0.52$, the margin of error is $E \approx 0.0490$ for $n = 400$.

(d) What happens to E when the sample size is four times as large, $n = 1{,}600$?

$E = 1.96\sqrt{\frac{(0.52)(0.48)}{1{,}600}} \approx 0.0245$**, half as large. The square root of** 4 **is** 2**.**

(e) Two surveys both find $\hat{p} = 0.52$, one with 100 people and one with $1{,}600$ people. Which 95% interval is wider?

The sample of 100**. A smaller** n **gives a larger margin of error.**

(f) Why not always take a very large sample?

A larger sample generally produces a more precise estimate, but it also costs more time and money.